

# Validity Analysis: MENTALCLOUDS

Target context: Global South Mental Health Practitioners — India-Anchored OMHC / Peer-Support Deployment

Overall risk: **HIGH**

## Deployment Context

**Use case:** A mental health professional or researcher deploys an LLM to automatically summarize counseling session transcripts. The system processes dialogues between therapists and clients, generating concise session notes that capture key therapeutic exchanges, client concerns, and counselor responses. The benchmark evaluates LLM quality on domain-specific summarization of sensitive mental health conversations.

**Target population:** Mental health professionals, clinical psychologists, and counselors in South Asia (India) and the broader Global South who work in online mental health community (OMHC) settings. Users are typically Hindi-English bilingual or English-speaking practitioners seeking AI-assisted documentation support.

## Dimension Scores

| Dimension                                                                                          | Score | Rating           | Confidence |
|----------------------------------------------------------------------------------------------------|-------|------------------|------------|
| Input Ontology    | 2     | Significant gaps | high       |
| Input Content     | 2     | Significant gaps | high       |
| Input Form        | 4     | Minor gaps       | high       |
| Output Ontology  | 2     | Significant gaps | high       |
| Output Content                                                                                     | 3     | Moderate gaps    | high       |
| Output Form     | 3     | Moderate gaps    | high       |
| Average                                                                                            | 2.7   |                  |            |

 = highest concern     = strongest dimension

## Dimension Key

| Abbr. | Dimension       | Definition                                                                                                   |
|-------|-----------------|--------------------------------------------------------------------------------------------------------------|
| IO    | Input Ontology  | Whether the benchmark's test case categories cover the query types expected in deployment.                   |
| IC    | Input Content   | Whether individual datapoint content is culturally and contextually appropriate for the target region.       |
| IF    | Input Form      | Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.             |
| OO    | Output Ontology | Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries. |
| OC    | Output Content  | Whether ground-truth labels align with the judgments of regional stakeholders.                               |
| OF    | Output Form     | Whether the expected output modality matches regional deployment needs and accessibility.                    |

## Overall Summary

MentalCLOUDS provides a methodologically thoughtful aspect-based summarization benchmark with a dual quantitative/qualitative evaluation design, English-only text input/output that matches the target deployment modality, and Indian-affiliated expert raters who provide partial alignment with the India-anchored OMHC/peer-support population. However, three HIGH-priority dimensions (Input Ontology, Input Content, Output Ontology) exhibit significant misalignment with the

deployment context. The benchmark's three-component taxonomy structurally omits suicide/self-harm risk flagging, family-intervention documentation, and spiritual/religious coping — content categories that elicitation, MHCA 2017 Section 115, and India's National Suicide Prevention Strategy 2022 establish as clinically and legally required. Session content is explicitly American-sourced from YouTube, with no compensating Global South counseling-summarization benchmark to triangulate against. The five expert raters are senior India-based clinicians with no peer-support, younger-cohort, or non-Indian Global South representation. Output form and input form are well-matched at the modality level, though ROUGE F1 underweights safety-critical content. Empirical evidence that LLMs systematically miscalibrate responses to suicidal ideation compounds these structural gaps. The benchmark is informative for evaluating CBT-arc summarization performance but cannot validate deployment-readiness for a system operating under Indian regulatory duty-of-care obligations.

## Practical Guidance

### What This Benchmark Measures

MentalCLOUDS measures LLM ability to produce aspect-based summaries of CBT-aligned counseling sessions across three components — symptom-and-history exploration, patient discovery, and reflective utterances [Q15, Q19] — using both lexical-overlap metrics (ROUGE/BERTScore) and a clinical-acceptability framework with hallucination identification [Q44, Q46]. For the target deployment, it provides a meaningful signal on the CBT-arc summarization core that the primary practitioner cohort cares about (input\_form: 4; partial strengths in output\_form: 3 and output\_content: 3), and acknowledges its own limitations transparently.

### Construct Depth

The construct of 'clinical-quality counseling summarization' is probed only partially. The three-component decomposition captures CBT-aligned reasoning steps, and the dual quantitative-qualitative evaluation [Q5, Q18] adds methodological depth. But the benchmark does not probe risk-flag recall, family-intervention documentation, spiritual/religious coping coverage, therapy-technique identification [Q63], or cross-cultural fidelity — all of which are required for the deployment's regulatory and clinical context. Expert qualitative evaluation is also limited to three models on 39 conversations [Q49, Q55], constraining statistical depth.

### What Else You Need

Supplementation should focus on the three HIGH-priority dimensions: (1) construct a deployment-specific risk-flag recall/precision evaluation set covering suicidal ideation, self-harm, and substance use, anchored to MHCA 2017 Section 115 obligations; (2) build a small India-anchored and broader Global South test corpus capturing family-system framing, spiritual/religious coping, and somatization patterns; (3) recruit a regional annotator pool that includes peer-support/OMHC counselors and non-Indian Global South practitioners to re-annotate a subset of model outputs and test for convergent-validity divergence from senior-clinician ratings; (4) replace or supplement ROUGE F1 with a content-coverage metric weighted toward safety-critical spans.

| ID  | Evidence                                                                                                                                                                                                  |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q5  | "The generated summaries are evaluated quantitatively using standard summarization metrics and verified qualitatively by mental health professionals." (p.1)                                              |
| Q15 | "Within each conversation, three pivotal counseling components (aspects) emerge – symptom and history exploration, patient discovery, and reflective utterances." (p.6)                                   |
| Q18 | "We undertake a comprehensive evaluation of the generated summaries across various architectures, employing a dual approach of quantitative and qualitative assessments." (p.9)                           |
| Q19 | "This section reports the aspect-based (psychotherapy element-based) summarization results based on the automatic evaluation scores." (p.9)                                                               |
| Q44 | "The evaluation framework encompasses six crucial parameters — affective attitude, burden, ethicality, coherence, opportunity costs, and perceived effectiveness." (p.12)                                 |
| Q46 | "Additionally, we incorporate a new parameter – the extent of hallucination. It is categorical – 0 (too much hallucinated), 1 (hallucination barely observed), and 2 (no hallucination observed)." (p.12) |

|            |                                                                                                                                                                                                                                    |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q49</b> | "We select the three best LLMs (MentalLlama, Mistral and MentalBART) for the expert evaluation based on the automatic result." (p.13)                                                                                              |
| <b>Q55</b> | "Out of the 39 test conversations, the majority of cases on average demonstrate no hallucination, with Mistral and MentalBART achieving 75.13% and 76.15% each, while MentalLlama shows a slightly higher value of 77.69%." (p.14) |
| <b>Q63</b> | "The models are also unable to identify psychotherapy types (e.g., cognitive behavior therapy) and therapy techniques, which form an integral part of counseling notes." (p.15)                                                    |